

FIZ1951-MÜHENDİSLİK İÇİN YARIİLETKEN FİZİĞİ DERSİ FORMÜL KAĞIDI

$$V = IR \quad J = \frac{I}{A} \quad R = \rho \frac{l}{A} \quad \rho = \frac{1}{\sigma} \quad J = \sigma E \quad E = \rho J \quad \vec{J} = \frac{\vec{I}}{A} = nq\vec{v}_s$$

$$\rho = \frac{1}{ne\mu} \quad \sigma = ne\mu$$

$$l = \vartheta_{ort} \tau_c \quad \mu_e = \frac{q\tau_c}{m^*} \quad \vartheta_d = \mu_e E \quad F = -eE$$

$$n = CT^{3/2} e^{-E_g/2kT}$$

$$C = \frac{2^{5/2} (m\pi k)^{3/2}}{h^3}$$

$$\sigma_n = ne\mu_n$$

$$\sigma_p = pe\mu_p$$

$$\sigma = \sigma_n + \sigma_p$$

$$\sigma = e(p\mu_p + n\mu_n)$$

$$n_i \cdot p_i = n_i^2 = p_i^2$$

$$n_i = p_i = \sqrt{n_i p_i}$$

$$N_{VB}(E) = \frac{1}{2\pi^2 \hbar^3} (2m_h^*)^{3/2} (E_V - E)^{1/2}$$

$$N_{CB}(E) = \frac{1}{2\pi^2 \hbar^3} (2m_e^*)^{3/2} (E - E_C)^{1/2}$$

$$f_n(E, T) = \frac{1}{1 + e^{\frac{E - E_F}{k_B T}}}$$

$$f_p(E, T) = 1 - f_n(E, T) \Rightarrow f_p(E, T) = \frac{1}{1 + e^{\frac{E_F - E}{k_B T}}}$$

$$n \cdot p = n_i^2$$

$$n_i = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right)$$

$$N_C = 2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2}$$

$$p_i = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right)$$

$$N_V = 2 \left(\frac{2\pi m_h^* k_B T}{h^2} \right)^{3/2}$$

$$E_g = E_C - E_V$$

$$n_i = \sqrt{N_C N_V} \exp\left(-\frac{E_g}{2k_B T}\right)$$

$$E_F = E_V + \frac{1}{2} E_g + \frac{k_B T}{2} \ln \frac{N_V}{N_C}$$

$$n_i = p_i = 2 \left(\frac{2\pi k_B \sqrt{m_e^* m_h^*}}{h^2} \right)^{3/2} T^{3/2} \exp\left(-\frac{E_g}{2k_B T}\right)$$

$$E_F = E_V + \frac{1}{2} E_g + 3 \frac{k_B T}{4} \ln \frac{m_h^*}{m_e^*}$$

$$p = p_i + N_A$$

N_A : : p tipi katkılı yarıiletkende
Akseptör atom konsantrasyonu
(1/m³)

$$N_A \gg n_i \rightarrow p = N_A$$

$$n = \frac{n_i^2}{N_A} \ll N_A$$

$$\sigma \approx N_A e \mu_p$$

$$n = n_i + N_D$$

N_D : : n tipi katkılı yarıiletkende
Donör atom konsantrasyonu (1/m³)

$$N_D \gg n_i \rightarrow n = N_D$$

$$p = \frac{n_i^2}{N_D} \ll N_D$$

$$\sigma \approx N_D e \mu_n$$

$$n = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right)$$

$$E_F = E_C + k_B T \ln\left(\frac{n}{N_C}\right)$$

$$p = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right)$$

$$E_F = E_V - k_B T \ln\left(\frac{p}{N_V}\right)$$

$$n = \frac{(N_D - N_A)}{2} + \sqrt{\left(\frac{(N_D - N_A)}{2}\right)^2 + n_i^2}$$

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$$\vec{E} = \frac{V}{l} \quad \mu_p = \frac{e\tau}{m_p^*} \quad \mu_n = \frac{e\tau}{m_n^*} \quad J_p = \sigma_p E = nq\mu_p E = pqv_s$$

Boşluklar için akım yoğunluğu

$$J_n = \sigma_n E = nq\mu_n E = nqv_s$$

Elektronlar için akım yoğunluğu

$$J = J_n + J_p = q(n\mu_n + p\mu_p)E$$

Genel akım yoğunluğu

$$J = \sigma E$$

$$J = e(p\mu_p + n\mu_n)E$$

$$\vartheta_{th} = \frac{l}{\tau_c}$$

$$J_n = qD_n \frac{dn}{dx}$$

$$J_p = -qD_p \frac{dp}{dx} \quad \frac{kT}{2} = \frac{m^* \vartheta_{th}^2}{2}$$

$$D_n = \mu_n \frac{kT}{q} \quad D_p = \mu_p \frac{kT}{q}$$

$$J_n = qn\mu_n E + qD_n \frac{dn}{dx}$$

$$J_p = qp\mu_p E - qD_p \frac{dp}{dx}$$

$$J_{toplam} = en\mu_n E + eD_n \frac{dn}{dx} + ep\mu_p E - eD_p \frac{dp}{dx}$$

$$V_{Hall} = \frac{IB}{nqt}$$

$20,10^{-3}$.

$$V_{Hall} = R_{Hall} \frac{IB}{t}, \quad R_{Hall} = \frac{1}{nq} \text{ veya } R_{Hall} = \frac{1}{pq}$$

$$L_z = m_l \hbar$$

$$\vec{\mu} = \left(-\frac{e}{2m}\right)\vec{L} \quad \vec{\mu}_s = \left(-\frac{e}{m}\right)\vec{S} \quad U_m = \vec{\mu} \cdot \vec{H}$$

Bohr manyetonu

$$U_m = m_l \left(\frac{e\hbar}{2m}\right) H$$

$$\mu_B = \frac{e\hbar}{2m}$$

$$U_m = \mu_B H m_l$$

$$\mu_B = 9.274 \times 10^{-24} \text{ J/T}$$

$$\vec{M} = \chi \vec{H} \quad \vec{M} = \frac{\vec{\mu}_{top}}{V}$$

$$\vec{B} = \mu_0 \vec{H} + \mu_0 \vec{M} = \mu_0 (1 + \chi) \vec{H} \quad \mu = \mu_0 (1 + \chi)$$

$$\vec{B} = \mu \vec{H}$$

- $\mu_r = \frac{\mu}{\mu_0}$ $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
 $B \rightarrow \text{Tesla, } 1\text{T} = 1\text{N/A.m}$
 $1\text{T} = 10^4 \text{ Gauss (G)}$
- $\chi = \frac{c}{T-T_c}$ • $\chi = \frac{c}{T}$

$$n^* = n_r - ik = \sqrt{\epsilon} , \quad n_r = \frac{c}{v}$$

$$\alpha = \frac{4\pi k}{\lambda} , \quad \alpha = A(h\nu - E_g)^\gamma$$

$\gamma = \frac{1}{2}$ direk bant aralıklı yarıiletkenlerde kullanılır

$\gamma = \frac{3}{2}$ indirek bant aralıklı yarıiletkenlerde kullanılır

$$R = \frac{(n_r - 1)^2 + k^2}{(n_r + 1)^2 + k^2}$$

$$T = (1 - R^2)e^{(-\alpha l)}$$

- $I = I_0 e^{(-\alpha l)}$

$$I_C = I_S(e^{V_{BB}/V_T} - 1), I_C \gg I_S \text{ şartı ile } I_C = I_S e^{V_{BB}/V_T} \text{ dir.}$$

$$I_S = \frac{A_E q D_n n_{p0}}{W}, \quad n_{p0} = n_i^2 / N_A$$

$$V_T = \frac{kT}{q}, V_T \cong 26 \text{ mV}$$

(oda sıcaklığında)

$$\beta = \frac{I_C}{I_B}, \quad I_B = \frac{I_C}{\beta}$$

$$I_E = I_C + I_B \quad \alpha = \frac{\beta}{\beta + 1} \quad \beta = \frac{\alpha}{1 - \alpha}$$

$$V_{R_B} = V_{BB} - V_{BE}, \quad V_{R_B} = I_B R_B$$

Dolayısıyla $I_B R_B = V_{BB} - V_{BE}$, ise $I_B = \frac{V_{BB} - V_{BE}}{R_B}$ olur.

$$V_{CE} = V_{CC} - V_{R_C}, \quad V_{R_C} = I_C R_C$$

$$V_{CE} = V_{CC} - I_C R_C, \text{ burada } I_C = \beta I_B$$

$$V_{CB} = V_{CE} - V_{BE}, V_{BE} \cong 0.7V \text{ dur, silikon transistör için.}$$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)^2 \quad I_D = K (V_{GS} - V_{GS(t)})^2$$

$$R = \left(\frac{n_y - n_\zeta}{n_y + n_\zeta} \right)^2 \quad \theta_k = \sin^{-1} \left(\frac{n_\zeta}{n_y} \right) \approx \frac{n_\zeta}{n_y}$$

$$\lambda = \frac{hc}{E} = \frac{1.24(eV)}{E(eV)} \mu m \quad \text{veya} \quad \lambda = \frac{hc}{E} = \frac{1240(eV)}{E(eV)} nm$$

, $c = \lambda \nu$

$$P_{\max} = V_{OC} I_{SC} FF$$

$$\eta = \frac{V_{OC} I_{SC} FF}{P_{in}} \quad FF = \frac{I_m V_m}{I_{SC} V_{OC}}$$

$$\epsilon_o = 8.85 \times 10^{-12} \frac{F}{m}, \quad \mu_o = 4\pi \times 10^{-7} \frac{N}{A^2}, \quad h = 6.63 \times 10^{-34} J.s$$

$$q = e = 1.6 \times 10^{-19} C, \quad k_B = 1.38 \times 10^{-23} J/K \quad m_o = 9.1 \times 10^{-31} kg$$